

142. A lens has a power of +2.0 Dioptre. Which one of the following statements about the lens is true ?

- (a) The lens is concave and has a focal length of 0.5 metre
- (b) The lens is convex and has a focal length of 2.0 metre
- (c) The lens is convex and has a focal length of 0.5 metre
- (d) The lens is concave and has a focal length of 2.0 metre

143. At nearly 70°C, sodium bicarbonate shows the property of gradually decomposing, which makes it usable in bakery products. The product of decomposition responsible for this use of sodium bicarbonate is

- (a) Carbon dioxide
- (b) Hydrogen
- (c) Water vapour
- (d) Oxygen

144. Number of molecules of water of crystallization in copper sulphate, sodium carbonate and Gypsum are

- (a) 5, 10 and 2 respectively
- (b) 10, 2 and 5 respectively
- (c) 5, 2 and 10 respectively
- (d) 2, 5 and 10 respectively

145. Which one of the following is the correct sequence of change in colours when a turmeric stain on white clothes is scrubbed by soap and then washed with water ?

- (a) Yellow – pink – blue
- (b) Yellow – reddish brown – yellow
- (c) Yellow – reddish brown – blue
- (d) Yellow – blue – pink

146. Which one of the following statements regarding Bleaching powder and D.D.T. is correct ?

- (a) Both are inorganic compounds
- (b) Both are organic compounds
- (c) Both contain chlorine
- (d) Both contain calcium

147. Which one of the following is the best example of desiccant ?

- (a) Silica gel
- (b) Polystyrene
- (c) Sodium chloride
- (d) Sodium carbonate

148. Which one of the following was the first mineral acid discovered ?

- (a) Sulphuric acid
- (b) Hydrochloric acid
- (c) Nitric acid
- (d) Phosphoric acid

149. संगलित स्फटिक (क्वार्टज़) का अपवर्तनांक 1.46 तथा नीलम का अपवर्तनांक 1.77 है। यदि क्वार्टज़ में प्रकाश की चाल v_q है और नीलम में प्रकाश की चाल v_s है, तो निम्नलिखित में से कौन सा एक संबंध सही है ?

(a) $v_q > v_s$

(b) $v_s > v_q$

(c) $v_s = v_q$

(d) $v_s = \frac{v_q}{2}$

150. एक अवतल दर्पण की स्थिति में, यदि किसी वस्तु को दर्पण के मुख्य फोकस F और ध्रुव P के बीच में रखा गया है, तो प्रतिबिम्ब के बारे में निम्नलिखित में से कौन सा एक कथन सही नहीं है ?

(a) प्रतिबिम्ब आभासी होगा

(b) प्रतिबिम्ब बढ़ा हुआ अथवा आवर्धित होगा

(c) प्रतिबिम्ब अनन्त पर बनेगा

(d) प्रतिबिम्ब सीधा होगा